Virus Model — Parameter Sweep

▼ About this sweep

This page sweeps the two parameters that most directly control epidemic dynamics in an SIR network model: **virusSpreadPercent** (transmission probability per contact per tick) and **recoveryPercent** (probability an infected node recovers per tick). Together they determine the **basic reproduction number R_0 = 6 × spread / recovery** — the expected number of new infections each infected node causes. When $R_0 < 1$ the virus fizzles; when $R_0 > 1$ it spreads through the network.

Two values of **gainResistancePercent** are compared — the probability that a recovered node gains permanent immunity rather than returning to susceptible. This parameter does not affect whether the virus spreads, only what fraction of the population ends up permanently immune vs. vulnerable when the epidemic ends.

The left chart plots **resistant %** vs R_0 on a log scale. Both curves follow a sigmoid: near zero below $R_0 = 1$, then rising steeply through the phase transition, leveling off at high R_0. Higher gainResistancePercent consistently produces more lasting immunity across all epidemic scenarios.

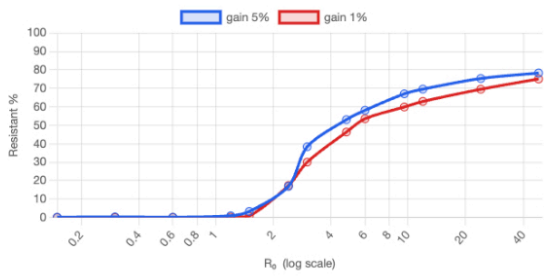
The right chart plots **ticks to completion** vs R_0. Both peak sharply just above $R_0 = 1$ — this is where the virus spreads just fast enough to persist but not fast enough to burn through quickly. Low gainResistancePercent produces a much higher and wider peak because recovered nodes cycle back to susceptible, giving the virus more fuel and extending the epidemic significantly.

Fixed: population = 150, averageNodeDegree = 6, initialOutbreakSize = 3, virusCheckFrequency = 1.

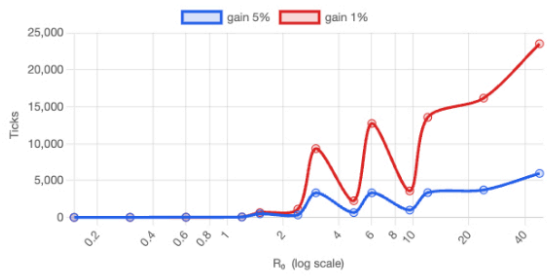
gainResistance A: 5% | gainResistance B: 1% | runs per combo: 10 | Run Sweep

Done — 50 combos × 10 runs = 500 total model runs.

Resistant % vs R_0



Ticks to finish vs R_0



gainResistance 5%

spread↓ / rec→	rec 1%	rec 2%	rec 5%	rec 10%	rec 20%
spread 0.5%	resist: 39.0% susc: 61.0% ticks: 4500 R_0≈3.0	resist: 3.3% susc: 96.7% ticks: 500 R_0≈1.5	resist: 0.2% susc: 99.8% ticks: 58 R_0≈0.6	resist: 0.5% susc: 99.5% ticks: 26 R_0≈0.3	resist: 0.1% susc: 99.9% ticks: 9 R_0≈0.1
spread 1%	resist: 57.0% susc: 43.0% ticks: 4355 R_0≈6.0	resist: 37.7% susc: 62.3% ticks: 2191 R_0≈3.0	resist: 1.1% susc: 98.9% ticks: 99 R_0≈1.2	resist: 0.2% susc: 99.8% ticks: 38 R_0≈0.6	resist: 0.1% susc: 99.9% ticks: 11 R_0≈0.3
spread 2%	resist: 69.5% susc: 30.5% ticks: 4207 R_0≈12.0	resist: 59.1% susc: 40.9% ticks: 2335 R_0≈6.0	resist: 20.7% susc: 79.3% ticks: 597 R_0≈2.4	resist: 1.1% susc: 98.9% ticks: 67 R_0≈1.2	resist: 0.1% susc: 99.9% ticks: 11 R_0≈0.6
spread 4%	resist: 75.6% susc: 24.4% ticks: 5052 R_0≈24.0	resist: 69.5% susc: 30.5% ticks: 2533 R_0≈12.0	resist: 53.1% susc: 46.9% ticks: 937 R_0≈4.8	resist: 17.3% susc: 82.7% ticks: 291 R_0≈2.4	resist: 0.6% susc: 99.4% ticks: 28 R_0≈1.2
spread 8%	resist: 78.3% susc: 21.7% ticks: 5971 R_0≈48.0	resist: 75.0% susc: 25.0% ticks: 2389 R_0≈24.0	resist: 67.0% susc: 33.0% ticks: 998 R_0≈9.6	resist: 52.9% susc: 47.1% ticks: 390 R_0≈4.8	resist: 12.6% susc: 87.4% ticks: 138 R_0≈2.4

gainResistance 1%

spread↓ / rec→	rec 1%	rec 2%	rec 5%	rec 10%	rec 20%
spread 0.5%	resist: 30.3% susc: 69.7% ticks: 12586 R_0≈3.0	resist: 0.7% susc: 99.3% ticks: 676 R_0≈1.5	resist: 0.0% susc: 100.0% ticks: 49 R_0≈0.6	resist: 0.0% susc: 100.0% ticks: 16 R_0≈0.3	resist: 0.0% susc: 100.0% ticks: 6 R_0≈0.1
spread 1%	resist: 52.9% susc: 47.1% ticks: 17248 R_0≈6.0	resist: 29.7% susc: 70.3% ticks: 6062 R_0≈3.0	resist: 0.1% susc: 99.9% ticks: 139 R_0≈1.2	resist: 0.1% susc: 99.9% ticks: 20 R_0≈0.6	resist: 0.0% susc: 100.0% ticks: 11 R_0≈0.3
spread 2%	resist: 62.7% susc: 37.3% ticks: 18038 R_0≈12.0	resist: 54.1% susc: 45.9% ticks: 8226 R_0≈6.0	resist: 20.2% susc: 79.8% ticks: 2026 R_0≈2.4	resist: 0.1% susc: 99.9% ticks: 29 R_0≈1.2	resist: 0.0% susc: 100.0% ticks: 11 R_0≈0.6
spread 4%	resist: 70.7% susc: 29.3% ticks: 22079 R_0≈24.0	resist: 63.0% susc: 37.0% ticks: 9101 R_0≈12.0	resist: 48.2% susc: 51.8% ticks: 2955 R_0≈4.8	resist: 15.5% susc: 84.5% ticks: 845 R_0≈2.4	resist: 0.3% susc: 99.7% ticks: 34 R_0≈1.2
spread 8%	resist: 75.1% susc: 24.9% ticks: 23532 R_0≈48.0	resist: 68.3% susc: 31.7% ticks: 10245 R_0≈24.0	resist: 59.9% susc: 40.1% ticks: 3591 R_0≈9.6	resist: 44.4% susc: 55.6% ticks: 1559 R_0≈4.8	resist: 16.2% susc: 83.8% ticks: 506 R_0≈2.4

Fixed: population=150, averageNodeDegree=6, initialOutbreakSize=3, virusCheckFrequency=1 — each cell = avg of 10 runs